MODULE 6: CONTEMPORARY TOPICS
(CO: 2)

Complete Study Textbook, Step-by-Step Derivations & VIT FAT Question Bank

Aligned with Official VIT BACHY101 Course Page Syllabus, Question Bank, and Past FAT Papers

Scope of This Textbook (Official VIT Syllabus - All Topics): (1) Contemporary Topics in Computational Chemistry, and (2) Contemporary Topics in Energy Devices or Functional Materials.

1 Contemporary Frontiers in Chemical Science & Engineering

1.1 Advanced Computational Chemistry: Machine-Learned Force Fields

Traditional computational chemistry faces a fundamental trade-off: Density Functional Theory (DFT) is highly accurate but scales cubically $O(N^3)$ with atom count, while empirical force fields are fast but lack electronic polarization. Machine-Learned Interatomic Potentials (MLIPs), such as MACE and NequIP, achieve DFT-level chemical accuracy at the computational speed of classical molecular dynamics, enabling atomic simulations of battery solid-electrolyte interphase (SEI) growth across millions of atoms.

1.2 Next-Generation Solid-State Lithium Metal Batteries

Current lithium-ion batteries using flammable liquid carbonate electrolytes suffer from thermal runaway risks and dendrite short-circuits. Solid-State Batteries replace the liquid with solid ceramic or polymer electrolytes:

- **Sulfides ($\mathbf{Li_{10}GeP_2S_{12}}$):** Ultra-high ionic conductivity ($> 10^{-2}$ S/cm at room temperature).
- **Garnet Oxides ($\mathbf{Li_7La_3Zr_2O_{12}}$):** Chemically stable against pure lithium metal anodes.
- Eliminating graphite in favor of pure Li metal boosts gravimetric energy density beyond 500 Wh/kg.

1.3 2D Functional Nanomaterials: MXenes

MXenes are 2D transition metal carbides, nitrides, or carbonitrides synthesized by selectively etching the "A" element from MAX phases using hydrofluoric acid (HF):

$$\mathrm{Ti_3AlC_2} + 3\ \mathrm{HF} \Rightarrow \mathrm{Ti_3C_2T}_x + \mathrm{AlF_3} + \frac{3}{2}H_2$$

where T_x represents surface terminations ($-O, -OH, -F$). MXenes combine metallic conductivity with hydrophilic surface chemistry, making them premier candidates for ultra-fast supercapacitors, EMI shielding in aerospace electronics, and water desalination membranes.